Multiplication properties of exponents

1 Write the following in expanded form:

a $3v^2u^5$

b $3m^4 \cdot 2n^2$

2 Write the following in exponential form:

a $a \cdot a \cdot b \cdot b$

b $3 \cdot 3 \cdot 3 \cdot 3 \cdot y \cdot y \cdot y$

c $3 \cdot u \cdot u \cdot u \cdot 5 \cdot v \cdot v \cdot v$

d $-3 \cdot c \cdot c \cdot b \cdot 5 \cdot c \cdot c \cdot a \cdot b$

e $(3p) \cdot (3p) \cdot (3p)$

3 Write the term that should go in the space to make the following statements true.

a $a^3 \cdot \square = a^6$

b $a^3 \cdot \square = a^4$

c $3x^{13} \cdot \square = 9x^{21}$

d $x^9y^8 \cdot \square = x^{11}y^{15}$

e $x^{16}y^6 \cdot \square = x^{22}y^{13}$

4 Simplify the following, giving your answer in exponential form:

a $2^2 \cdot 2^3$

b $3^9 \cdot 3^{10}$

c $y^2 \cdot y^6$

d $x^4 \cdot 10x^3$

e $4y^5 \cdot 3y^2$

f $4y^3 \cdot 6y$

g $8y^9 \cdot 5y^7$

h $8y^4 \cdot 8y^3$

i $9m^2 \cdot 6m^2$

j $7y^3 \cdot 5y^4$

k $u^2 \cdot u^6 \cdot u^3$

l $6y^7 \cdot 6y^5 \cdot 6y^3$

m $4y^8 \cdot 6y^6 \cdot 3y^3$

n $8y^3 \cdot 9y^4$

o $4y^2 \cdot 5y^4 \cdot 6y^8$

p $3y^7 \cdot 5y^8 \cdot 2y$

5 Simplify the following, giving your answer in exponential form:

a $5u^4v^2 \cdot 8u^3v^4$

b $6bc^4 \cdot 8b^6a \cdot 0$

c $8m^2n^4 \cdot 7m^6n$

d $8wuv \cdot 4uwv \cdot 7uvw \cdot 6uv \cdot 6vu$

6 Simplify the following, giving your answer in exponential form:

a $9y^9 \cdot 8\left(-y\right)^8 \cdot 7y^7$

b $4y^4 \cdot \left(-2y^6\right) \cdot 4y^5$

c $\left(-5y^6\right) \cdot \left(-3y^7\right) \cdot \left(-4y^7\right)$

d $\left(-9y^5\right) \cdot \left(-8y^9\right)$

e $\left(-9b^4\right) \cdot 4b^4$

f $\left(-7y^5z\right) \cdot \left(-3yx^3\right)$

g $2v^2w \cdot \left(-5u^2v^3\right) \cdot 3u^3w^4$

h $\left(-4y^7\right) \cdot \left(-6y^2\right) \cdot \left(-4y^2\right)$

i $\left(-8q^4\right) \cdot p^2 \cdot \left(-7q^4\right) \cdot p^3$

7 Simplify the following, giving your answer i[redacted]onential form:

a $\left(j^2\right)^5$ b $\left(c^9\right)^2$ c $\left(f^8\right)^6$ d $\left(w^3\right)^4$

e $y^8 \cdot (3y)^5$ f $\left(u^6\right)^3 \cdot u^2$ g $\left(w^9 v^6\right)^4$ h $\left(k^5\right)^3 \cdot \left(k^2\right)^8$

i $6\left(r^6\right)^5 \cdot 4\left(r^4\right)^7$ j $\left(x^9 y\right)^7 \cdot \left(xy^2\right)^4$ k $10\left(r^7\right)^5 \cdot (2rs)^4$ l $\left(p^5\right)^9 \cdot \left(q^6\right)^{10}$

m $\left(p^{10}\right)^6 \cdot \left(p^4\right)^3$ n $\left(\left(x^3\right)^4\right)^5$

8 The length of the base of a triangle is $3c^4$ and the length of its perpendicular height is $6c^8$. What is the expression for the area of the triangle?

9 What quantity multiplied by itself results in w^{100}?

10 Consider $\left(r^2\right)^4$.

a State whether the following expressions are equivalent to $\left(r^2\right)^4$:

i $r^2 \cdot r^4$ ii $(r \cdot r) \cdot (r \cdot r \cdot r \cdot r)$

iii $(r \cdot r)^4$ iv $(r \cdot r) \cdot (r \cdot r) \cdot (r \cdot r) \cdot (r \cdot r)$

v $r^2 \cdot r^2 \cdot r^2 \cdot r^2$

b State whether the following statements are correct.

i $\left(r^2\right)^4 = r^{2+4}$ ii $\left(r^2\right)^4 = r^{2 \cdot 4}$

c Fill in the box to complete the rule: $\left(r^2\right)^4 = r^{\square}$

11 Consider the following terms which form a pattern: $\left(x^2\right)^1, \left(x^2\right)^2, \left(x^2\right)^3, \left(x^2\right)^4, \ldots$
What would the fifth term be? Fully simplify your answer.

12 Simplify, the following expressions:

a $\left(-x^3\right)^4$ b $\left(4y^4\right)^3$ c $\left(8y^6\right)^2$ d $\left(3y^6\right)^2$

e $\left(-2x^2\right)^2$ f $\left(-3p^2\right)^5$ g $\left(2u^5 v^5\right)^4$ h $\left(2r^2 s\right)^4$

13 Write the expression that should go in the parentheses to make the following statements true.

a $(\square)^2 = q^{12}$ b $(\square)^3 = 27q^{12}$ c $(\square)^2 = 9p^8 q^{18}$